

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 08 · LECTURE NOTES

RNA Structure

Dynamic programming on intervals, and a model that is honestly incomplete.

MODULE OUTCOMES

- Explain the folding problem and the base-pairing model.
- Derive the Nussinov recurrence.
- Explain energy minimisation and why it is better.
- Explain pseudoknots and why they are excluded.
- State what a predicted structure establishes.

1 The problem

RNA FOLDS BACK ON ITSELF, AND THE SHAPE DETERMINES WHAT IT DOES

Complementary bases pair — A with U, G with C, and G with U more weakly — and a single strand **pairs with itself**, producing stems (stacked pairs), hairpin loops, internal loops, and bulges.

Which matters because structure is function — for ribosomal and transfer RNA, for regulatory elements in untranslated regions, and for the many non-coding RNAs whose entire role is structural rather than informational.

And structure is conserved where sequence is not — compensatory mutations preserve a pair while changing both of its bases (a G-C becomes an A-U) — **which is evidence of function that plain sequence comparison misses entirely**, and which §4 exploits.

So the computational problem is: given a sequence, predict the set of pairs — **which is a combinatorial optimisation with a physically motivated objective**, and is therefore unusually well posed for this field.

2 Nussinov

$$N(i, j) = \max \{ N(i+1, j), \quad N(i, j-1), \quad N(i+1, j-1) + \delta(i, j), \quad \max_k [N(i, k) + N(k+1, j)] \}$$

- The subproblem is an interval $i..j$ rather than a pair of prefixes — **which is the structural difference from Module 02**, and is what makes the fill order by increasing interval length rather than by row.

- **The first three cases: leave i unpaired, leave j unpaired, or pair i with j** (with $\delta = 1$ if those two bases can pair and 0 otherwise) — **which is the max-of-cases structure you already know.**
- **The fourth case is bifurcation: split the interval at some k and add the two halves' scores** — **which is what makes the algorithm $O(n^3)$ rather than $O(n^2)$** , since it introduces a loop over k .
- **The bifurcation case is what makes it cubic, and it is the case most often omitted in a first implementation** — producing an algorithm that cannot represent two independent stems side by side, which is a quiet and plausible-looking failure. **Intervals rather than prefixes: that is the whole structural difference.**

AND MAXIMISING PAIR COUNT IS THE WRONG OBJECTIVE, WHICH THE NEXT MODEL FIXES

Pairs do not contribute equally to stability — stacked pairs within a helix are strongly stabilising and isolated pairs are essentially not — so counting pairs rewards structures that would never form, scattering single pairs across the molecule.

So Zuker's algorithm minimises free energy instead, with experimentally measured parameters for stacking, for loops of each type and size, and for bulges — the same interval decomposition with a much better objective, and more cases to handle.

Which is a good example of the pattern this course keeps meeting: the algorithm was the easy part and the scoring function was the research — exactly as in Module 03 §1, where the recurrence was 1970 and the matrices were two decades of work.

And the energy parameters are measured in the laboratory, which makes the prediction physically grounded rather than merely consistent — **within the model's assumptions, which §3 names explicitly** because they are load-bearing.

3 What the model excludes

- **No pseudoknots** — pairs may not cross (if i pairs with j and k with l , then either both of k, l lie inside $i..j$ or both lie outside) — **which is exactly what makes the interval decomposition valid**, and **pseudoknots exist and are functionally important**, including in several viral elements.
- **Allowing them in general makes the problem NP-hard, so the exclusion is a real algorithmic constraint rather than laziness** — and specialised heuristics handle restricted pseudoknot classes at higher cost.
- **One sequence, one structure** — whereas real RNA interconverts between conformations, and **some regulatory elements work precisely by switching** between two structures in response to a ligand or a temperature change, which the model cannot represent at all.
- **No ligands, no proteins, no ions** — and **magnesium in particular stabilises tertiary structures that the model cannot see**, so predictions for structured RNAs in their biological context are systematically incomplete.
- **And the minimum free energy structure is frequently not the one that forms** — the energy landscape is shallow and folding is kinetic — **which is why the partition function and the resulting base-pair probabilities are the better thing to report** than a single structure (McCaskill's algorithm computes them with the same decomposition and sum in place of max, exactly as Module 07 §1's forward relates to Viterbi).

4 Doing better

COMPARATIVE FOLDING BEATS SINGLE-SEQUENCE FOLDING, BECAUSE CONSERVATION IS EVIDENCE THE ENERGY MODEL IS NOT

If several related sequences fold to the same structure, and compensatory mutations preserve the pairs while changing the bases, that is independent evidence that the structure is real — **which the thermodynamic model does not and cannot provide**, since it would predict a structure for random sequence too.

So the better methods fold an alignment rather than a single sequence, combining thermodynamic scores with a covariation term — **and they are measurably more accurate** on benchmarks of known structures, by a wide margin for structured families.

Which is the same move as Module 07 §3's profile HMMs: the family carries information that no individual member does — in a second setting, and it is worth noticing as a general principle rather than two tricks.

And the honest claim is: 'the minimum free energy structure under this parameter set, with these base-pair probabilities' — not 'the structure of this RNA', which requires an experiment (Module 01 §4's no-ground-truth position, and Module 13's claim form).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Durbin et al., chapter 10 (library copy)	§§2 and 4 — RNA structure, including stochastic context-free grammars, which is the unifying framing.
Zuker — Mfold web server for nucleic acid folding (free)	§2's energy minimisation — and the server is worth running on a sequence you care about.
Lorenz et al. — ViennaRNA Package 2.0 (free)	§§3 and 4 — partition function, base-pair probabilities, and comparative folding, as working software.
Mathews et al. — the nearest-neighbour energy parameters	§2's measured parameters — where the numbers come from, which is laboratory work rather than computation.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Pair a short sequence by hand** and draw the structure.
- 2 **Write the Nussinov recurrence** with all four cases.
- 3 **Implement it**, filling by increasing interval length.
- 4 **Omit the bifurcation case** and find a sequence where it matters.
- 5 **Compare Nussinov's output to an energy minimiser's** on the same sequence.
- 6 **Construct a pseudoknot** and confirm the recurrence cannot represent it.
- 7 **Fold a known tRNA** and compare to its published structure.
- 8 **Compute base-pair probabilities** and compare to the single MFE structure.
- 9 **Fold an alignment of a conserved family** and compare to folding one member.
- 10 **Find a compensatory mutation** in that alignment.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does RNA structure matter, and what is conserved?
- 2 What is the subproblem, and how does that differ from alignment?

- 3 Give the four cases, and say which makes it cubic.
- 4 Why is maximising pairs the wrong objective?
- 5 What replaced it, and where did its parameters come from?
- 6 What general pattern does that illustrate?
- 7 State five assumptions the model makes.
- 8 Why are pseudoknots excluded?
- 9 Why report base-pair probabilities?
- 10 Why does comparative folding work better?